

Summarized chain of thought (Ordinary NP-Hardness at the Basic Semidefinite Threshold)

OpenAI

Original prompt (excerpts)

Problem

Prove or refute the following ordinary NP-hardness assertion for the basic-SDP approximation threshold of every fixed finite Max-CSP. The intended improvement is to obtain matching NP-hardness without the Unique Games Conjecture. The deterministic inapproximability consequence, and the interpretation of an algorithmic counterexample, use the assumption $P \neq NP$.

FIXED LANGUAGE AND FINITE INPUTS. Fix a nonempty finite alphabet D and a finite nonempty collection Γ of predicates $f: D^{k_f} \rightarrow \{0,1\}$, with positive integer arities k_f . The alphabet, all predicate truth tables, and $k = \max_f k_f$ are fixed independently of the input size; they are not growing-alphabet or growing-arity parameters.

An instance I has finitely many variables $x_1, \dots, x_n$ in D and a finite nonempty list of constraint occurrences $e=1, \dots, m$. Discard unused variables and relabel the variables that occur as $x_1, \dots, x_n$; thus n is at most the total number of argument positions in the listed constraints. Occurrence e specifies f_e in Γ , an ordered tuple $(v_e(1), \dots, v_e(k_e))$ of variable indices, and a positive rational weight w_e , with $\sum_e w_e = 1$. Variable indices may repeat within a tuple, and different occurrences may have the same tuple or the same underlying scope. Rational numerators and denominators and variable indices are encoded in binary. This weighted input convention is part of the problem; zero-weight occurrences can be discarded before normalization. Unweighted instances are the special case $w_e = 1/m$.

For an assignment $a: \{1, \dots, n\} \rightarrow D$ define

$\text{val}_I(a) = \sum_e w_e f_e(a(v_e(1)), \dots, a(v_e(k_e)))$,

and define $\text{OPT}(I) = \max_a \text{val}_I(a)$. Thus the objective is the normalized weight of satisfied predicates. There is no promise that all constraints can be simultaneously satisfied.

THE BASIC SDP. Use the following specific relaxation, in the vector/local-distribution form of Raghavendra's SDP(I) and its standard BasicSDP formulation. Let S_e be the set of distinct variable indices appearing in occurrence e . Introduce real vectors u_0 and $u(v, d)$, for v in $\{1, \dots, n\}$ and d in D . Introduce a separate probability distribution μ_e on D^{S_e} for every constraint occurrence e . Here β in D^{S_e} means a map $\beta: S_e \rightarrow D$. Maximize

$$\sum_e w_e \sum_{\beta \in D^{S_e}} \mu_e(\beta) f_e(\beta(v_e(1)), \dots, \beta(v_e(k_e)))$$

subject to all of the following:

$\text{dot}(u_0, u_0) = 1$;

$\sum_{d \in D} u(v, d) = u_0$ for each variable v ;

$\text{dot}(u(v, d), u(v, d')) = 0$ for $d \neq d'$;

$\mu_e(\beta) \geq 0$ and $\sum_{\beta} \mu_e(\beta) = 1$ for every e ;

$\text{dot}(u(v, d), u(w, d'))$

$= \sum_{\beta: \beta(v)=d, \beta(w)=d'} \mu_e(\beta)$

for every e, v, w in S_e and d, d' in D , including $v=w$;

$\text{dot}(u(v, d), u_0)$

$= \sum_{\beta: \beta(v)=d} \mu_e(\beta)$

for every e, v in S_e and d in D .

Here dot denotes the real inner product. The vectors have a common finite-dimensional real inner-product space; equivalently their Gram matrix is positive semidefinite. Dimension $n|D|+1$ suffices to express all such Gram matrices. The value $\text{BasicSDP}(I)$ is the maximum of this relaxation. No higher-round consistency or other strengthening is added.

Repeated positions refer to the same $\beta(v)$, not independently sampled copies. Even when $S_e = S_{e'}$, the distributions μ_e and $\mu_{e'}$ are separate: they must agree with the shared vectors on the specified singleton and pairwise marginals, but are not required to agree on their entire joint distributions. This convention fixes the relaxation even for duplicate scopes.

RATIO THRESHOLD. Use the reward-ratio convention, in which larger ratios are better, and define the real constant

$\alpha_Gamma = \inf \{ \text{OPT}(I) / \text{BasicSDP}(I) : I \text{ is an instance over } \Gamma \text{ and } \text{BasicSDP}(I) > 0 \}$.
Set $\alpha_Gamma = 1$ if this set is empty, in particular when all allowed predicates are identically false. Zero-SDP instances are excluded from the quotient; their objective does not create a division-by-zero convention. This constant lies in $[0, 1]$. It is a threshold: the problem does not require exact attainment by a single rounding algorithm or a uniform algorithm for computing α_Gamma .

ASSERTION TO ESTABLISH. For every fixed D, Γ as above and every fixed rational number r with $\alpha_Gamma < r < 1$,

there exist rational constants s, c with

$0 \leq s < c < 1$ and $s < r < c$,

such that the following promise problem is NP-hard under deterministic polynomial-time many-one reductions from Boolean satisfiability:

YES: $\text{OPT}(I) \geq c$;

NO: $\text{OPT}(I) \leq s$.

A reduction must map every satisfiable Boolean formula into the YES set and every unsatisfiable formula into the NO set, using only instances of this same fixed Γ and the stated weight encoding. Behavior outside the two promised sets is unrestricted. The reduction, its polynomial bound, and c, s may depend on D, Γ, r ; they are fixed before the input formula size grows.

This is a single worst-case multiplicative-ratio assertion. It does not require hardness at every point of an SDP approximation curve, for every SDP-gap witness, or with perfect completeness $c=1$. The strict slack in r and the existence of some separated constants c, s are intentional. For $\alpha_Gamma = 1$ there is no r in the stated interval and no hardness claim is required. If $\alpha_Gamma < 1$, an exact polynomial-time optimizer would also achieve every $r < 1$, so a separate endpoint assertion for $r=1$ is unnecessary.

COMPLEXITY INTERPRETATION. Such a gap reduction excludes, under $P \neq \text{NP}$, a deterministic polynomial-time algorithm that returns an assignment of value at least $r \cdot \text{OPT}(I)$ on every instance: YES instances would yield a verifiable value above s and NO instances could not. No claim excluding bounded-error randomized approximation algorithms follows here from $P \neq \text{NP}$ alone, and no stronger randomized-complexity hypothesis is silently imposed.

A complete affirmative resolution establishes the displayed NP-hardness statement for every fixed language and every indicated r . A complete negative resolution rigorously refutes that universal hardness assertion under the stated $P \neq \text{NP}$ interpretation. For example, a fixed language together with a deterministic polynomial-time algorithm and a proved guarantee $\text{val}_I(a) \geq \beta \cdot \text{OPT}(I)$ for all its instances, for some constant $\beta > \alpha_Gamma$, would suffice under $P \neq \text{NP}$. A resolution is not required to prove the assumed separation $P \neq \text{NP}$ itself. Failure of one rounding rule or one attempted reduction is not a negative resolution. No proof or candidate algorithm is supplied here.

1 From the BasicSDP threshold to projection leakage

The assistant first checks whether repeated variables or occurrence-specific local distributions weaken the relaxation enough to admit a counterexample. The assistant finds no apparent escape through those conventions in Raghavendra's optimal CSP approximation theorem [50]. Its dependence on the Unique Games Conjecture (UGC) instead motivates a proposed converse. Alphabet-dependent rounding bounds for Unique Games [17] constrain one attempted baseline construction; the finite-valued CSP

dichotomy [55] also gives the assistant no tractable-language loophole.

The key distinction is between a small soundness/completeness ratio and completeness near one. The assistant observes that the Khot–Minzer–Safra 2-to-2 Games theorem [39] does not directly supply the latter. The assistant first considers the Goemans–Williamson near-satisfiable Max-Cut guarantee [29] and the odd-cycle obstruction to strong repetition [52]. A baseline gadget then adds three dummy labels, guaranteeing value at least $T = 1/3$ while retaining a Khot–Vishnoi SDP gap [38]. If the universal assertion forces hardness close to this baseline, the assistant argues that completeness must approach one:

VERBATIM EXCERPT

Wait insight: baseline T constant and hardness YES near 1 NO $\leq T + o(1)$ is already enough for UGC via parallel repetition! NO needs val bounded from 1, not tiny, then repeat constant power depending δ alphabet.

The assistant invokes Rao’s alphabet-independent projection-game repetition bound [51], leaving quantitative parameters as a further step in the proposed equivalence. For an algorithmic alternative, the assistant recalls constant-round sum-of-squares treatment of noisy-cube gaps [7] and contrasts the Arora–Barak–Steurer subexponential algorithm [1] with the desired polynomial running time. Results for certified small-set expanders and globally hypercontractive graphs [5, 4] prompt the objection that arbitrary constraint graphs need not satisfy those assumptions. It considers extending Trevisan’s spectral partial-assignment method [56] and uses the Khot–Kindler–Mossel–O’Donnell reduction [40] to motivate an affine-game approach, while questioning the precise field formulation. Hashing and coordinatewise Boolean recovery encounter accumulating errors or inconsistent cluster shifts.

Attention then turns to hardness gadgets. The Rich 2-to-1 equivalence [16] suggests hiding the matching structure. The assistant examines the Dinur–Khot–Kindler–Minzer–Safra composition [21] and Minzer–Zheng’s alphabet–soundness tradeoff [46] as possible outer constructions, and invokes Håstad’s near-one versus one-half parity-equation hardness [32]. Using the shortcode framework of Khot–Minzer–Safra and Barak–Kothari–Steurer [39, 8], the assistant explores a matrix test and a tensor extension whose honest acceptance can rise from one-half to three-quarters. Yet projected tensor queries can reveal the outer constraint: an answer may adapt to that constraint instead of encoding a fixed label. Restricting monomials to individual blocks and using sparse noise fails to hide recognizable duplications:

VERBATIM EXCERPT

Thus at most one informative i . So cheater chooses arbitrary $g_{\{v_i\}}$, on exact duplicated input matches and passes, independent of actual satisfiability!

Dense noise obstructs this response but loses the desired completeness. The assistant therefore retains the proposed Unique Games route while turning to the problem of hiding projections without sacrificing completeness.

2 Hidden structure and the search for a stability gap

The proposed Unique Games route to the original BasicSDP-threshold assertion repeatedly confronts one requirement: honest assignment encodings must survive perturbations much better than arbitrary answers. Tensor products make honest errors rare, but low-rank representations reveal projected coordinates or recoverable constants. A prover can exploit that information without decoding an assignment. Availability flags for selecting successful constraints also encounter a counting obstruction from uniqueness. These attempts repeatedly return to the affine 2-to-2 and inverse-shortcode framework [39, 8].

After considering Khot and Moshkovitz’s real-code candidate route [41], the assistant turns to Subspaces Near-Intersection in Eldan and Moshkovitz’s *Reduction From Non-Unique Games To Boolean Unique Games* as an alternative starting point [25]. Gaussian queries offer small geometric perturbations, but nonlinear Lipschitz responses can be similarly stable:

VERBATIM EXCERPT

So stability comparable to honest. Thus noise-based Gaussian tester fails because low-degree linear dependence not enforced; need linearity test via rank-one transformations that preserves F_x more than smooth functions.

Compressed tensors with independent linear masks then offer a possible repair. Fourier eigenvalues suggest how rank-one frequencies might encode assignments, but the assistant cannot yet establish consistency across projected questions. The assistant consults Minzer and Zheng’s *Near Optimal Alphabet-Soundness Tradeoff PCPs* for its composed construction and subspace identifications [46]. A BLR-style linearity step is proposed to extract aligned labels [13]. It also considers Friedgut’s junta theorem for extracting a small set of relevant coordinates, but finds that sparse-noise control does not constrain the singular padded inputs used by projection tests [28].

An amplification insight changes the target: if undesirable frequencies lose increasingly more than honest encodings per step, repeated noise might create constant dishonest loss while honest loss vanishes. The assistant does not find a suitable higher-degree walk. Recoverable components defeat masking, and majority blocks are less stable than competing dictators under sparse noise. The Kasami–Tokura low-weight Reed–Muller classification guides a search for structured, highly biased multilinear forms [36]. A more abstract equivariant-decoder proposal would still need inverse-shortcode decoding and adaptation of the Dinur–Khot–Kindler–Minzer–Safra outer construction [21].

The assistant also sketches a conditional connection between the BasicSDP threshold and UGC, using the Khot–Vishnoi construction [38] and Raghavendra-style dictatorship tests [50]. The gap calculation uses Bonami–Beckner hypercontractivity [14, 12]; Rao and Dinur–Steurer projection-game repetition are considered for alphabet-independent amplification [51, 22]. For the dictatorship-test transfer, the assistant invokes the low-influence multilinear invariance principle of Mossel, O’Donnell and Oleszkiewicz [47].

A final proposal uses rounding on a long cyclic group $\mathbb{Z}_M$: small increments rarely cross interval boundaries. The assistant considers Khot and Moshkovitz’s hardness result for real linear equations as a possible starting point [42]. Exact integer equations appear to improve projection hiding, while incompatible winding numbers on local tori might force cheating functions to lose stability. The assistant identifies a difficulty with smoothing: it may leave exceptional regions where outputs cease to be circle-valued. It leaves extracting winding numbers and controlling correlated strategies as the next obstacles to obtaining the desired completeness–soundness separation.

3 From winding obstructions to global consistency

The proposed Unique Games route toward the BasicSDP threshold encounters a recurring problem: local consistency does not supply one global labeling. A cyclic construction based on scalar winding appears vulnerable to vector-valued maps into higher-dimensional spheres. Tensoring these maps and projecting through a shared Gaussian could correlate their singularities and achieve the honest strategy’s error scale. The assistant considers this a possible failure mechanism for the cyclic construction.

The assistant therefore explores an algorithmic alternative for nearly satisfiable affine Unique Games, with constraints $x_u + x_v = c_e$ over $\mathbb{F}_2^\ell$. The desired approximation must remain useful as the alphabet grows. The assistant invokes the Khot–Vishnoi noisy-cube construction [38] against a universal equivariant rounding map. Separately solving Boolean character projections by Goemans–Williamson

rounding [29] leaves their errors unsynchronized; graph squaring compounds error instead of eliminating it. The conditional hardness route continues to use Raghavendra’s framework [50] and Rao’s alphabet-independent projection-game repetition [51].

The assistant also recalls Khot–Saket hierarchy gaps [43] while assessing stronger relaxations. The algorithmic search contrasts parity and random-CSP hierarchy lower bounds associated with Grigoriev and Schoenebeck [31, 54] with the sum-of-squares hypercontractive refutation of noisy-cube and short-code gaps [7]. Algorithms for Unique Games on expanders [2] and the subexponential algorithm of Arora, Barak and Steurer [1] motivate graph decomposition, while the condition-and-round method for certified small-set expanders [5] motivates a collision potential for differences of two pseudolabelings. The assistant does not obtain the desired alphabet-uniform rounding from these approaches.

Alpha-expansion graph cuts [15] encounter the unknown gauge of shifted Potts constraints. Connection-Laplacian Cheeger inequalities [6] suggest partial synchronization, but continuous rotations still need conversion into discrete group labels. A separate proposed hardness route would combine a hypothetical stable coloring with the Khot–Minzer–Safra inverse framework [39]; the assistant leaves the coloring construction unresolved.

Shared error locations suggest interleaved-code decoding, but syndrome rank can remain small while the number of errors grows. Another proposal geometrically separates bad edges while charging progressively smaller costs to good edges.

Its embedding, however, depends on the unknown true labels. The assistant considers Papadimitriou–Yannakakis expander degree reduction [49], splitting variables into copies joined by equality constraints. Forest compression then suggests reducing the task to finding linearly many compatible cycles beyond a spanning forest. The assistant leaves both that solver and the required deterministic sampling argument unresolved, and uses high-girth examples to examine limitations of local algorithms.

Barto and Kozik’s robust satisfiability theorem [11] offers a fixed-language comparison, but the assistant questions whether the relevant constants can be made alphabet independent. Orthogonal separators [19] similarly expose alphabet-dependent losses in the proposed collision-based clusters. The assistant also considers using Barak–Kelner–Steurer sum-of-squares rounding [9] to extract a high-norm moment matrix, then encounters dimension-dependent conditioning costs. A more concrete attempt conditions semidefinite pseudolabels on random linear-hash fibers. Surviving residuals seem less frequent, but the restricted vectors no longer sum to a common reference vector:

VERBATIM EXCERPT

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Missing mass could be filled, but to shrink p need *renormalize* near intersection and **realign**
\ (u'_0(v)).
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Repairing this defect may restore the discarded error. Holenstein’s correlated sampling [33] offers a coupling of fiber-conditioned labels, but its mismatch cost also threatens the proposed error reduction. Finally, countably many approximately consistent character projections do not ensure a coherent measurable labeling: their common kernel and dependence on the selected lift remain obstacles. BLR linearity testing [13] would decode jointly additive scalar potentials, but the assistant has not obtained that joint coherence. The assistant ends this exploration with the global consistency obstacles unresolved and continues searching for an alphabet-independent algorithm or a route to the BasicSDP threshold.

4 Nonlinear decoders and the amplification barrier

The attempt to match the BasicSDP threshold for unconditional hardness continued through a proposed Unique Games reduction. Honest encodings needed to withstand noise while inconsistent answers became decodable or unstable. Tensor constructions made honest error exponentially small by using rare gates, but the same gates protected arbitrary functions of a surviving matrix channel:

VERBATIM EXCERPT

Thus soundness gap $\approx \epsilon$, not enough. Need ensure dummy axes themselves hard so no arbitrary gating.
Imbalanced cycle obstacle.

Reconsidering the Khot–Minzer–Safra construction [39] focused attention on representations compatible with many projections and noise that disrupts adaptive choices among them. The assistant revisited the outer reduction of Dinur, Khot, Kindler, Minzer and Safra [21] and Minzer–Zheng’s equivalence-class construction [46], and inspected the Barak–Kothari–Steurer shortcode formulation [8]. Rich pairings [16] suggested scrambling labels, but fixed permutations preserved the obstruction and edge-dependent permutations exposed it. Smooth-code alternatives prompted an analogy with Khot–Moshkovitz’s Boolean noise test [41]; Gowers–Hatami stability [30] was another comparison, without a demonstrated application to the proposed nonuniform directions. Nonlinear equivariant decoders became the main alternative. The target was an unbounded separation between nonlinear stability and the best compatible linear decoding, sufficient for the proposed reduction.

Two difficulties then evolved together. Projection aliases Fourier frequencies, so low rank need not yield agreeing decoded labels. A proposed inclusion–exclusion formula bounded Fourier truncation independently of the number of outer coordinates, but controlling projected queries and extracting consistent labels remained unresolved. Meanwhile, concatenating nonlinear decoders allowed linear competitors to split into separate scalar lifts. Recalled Grassmann hypercontractivity of Ellis, Kindler and Lifshitz [26] motivated seeking stronger rank bounds. Dinur’s gap amplification [23] suggested exploiting structured completeness errors; Dumer–Micciancio–Sudan code-distance hardness [24] suggested sparse outer solutions. Neither idea removed the projection obstacle, which the assistant contrasted with the uniform query marginals in Håstad’s Max3Lin reduction [32]. Generic OR amplification encountered equality’s transitivity, and sparse-column encodings exposed their projections through duplicated entries.

Perfect colorings suggested decoders whose output characters occupy a single Fourier degree. The Nisan–Szegedy degree bound [48] limited the number of relevant Boolean coordinates, prompting a distinction between Boolean degree and degree over larger alphabets. Writing p for nonlinear noise failure and $h(r)$ for the least noise cost of simultaneous linear lifts of r independent logical characters, the desired advantage was a growing ratio $h(r)/p$. A simplex-mixing proposal sought to combine small decoders through symmetric injections into a larger logical space, preserving an existing advantage while enlarging the output. Recursive amplification again met fragmentation:

VERBATIM EXCERPT

But after many levels rank may collapse along branches; fixed b implies stabilization at $\text{factor} \leq b$.
Need increasing b or avoid collapse via mixing outer layers.

Symmetry-based sections still admitted componentwise cheating, while permutation-valued path encodings could reveal projection pairings through a single comparison. The assistant retained simplex mixing as a way to preserve an existing advantage, while continuing to seek amplification and a soundness argument that would turn the decoder separation into the requested BasicSDP hardness reduction.

The assistant also retained a conditional fallback: combine the Khot–Vishnoi gap [38] with Raghavendra’s dictatorship-test framework [50] and the Mossel–O’Donnell–Oleszkiewicz invariance principle [47], using Rao’s or Dinur–Steurer’s alphabet-independent repetition [51, 22] for the reverse implication to Unique Games. Its algorithmic alternative recalled the condition-and-round approach of Bafna and collaborators [5], while continuing to confront the small-set expansion barrier.

5 From coordinated-error barriers to a candidate decoder gap

The assistant revisited Raghavendra’s conditional CSP theorem and the Khot–Vishnoi gap [50, 38], and considered improving Goemans–Williamson rounding as an algorithmic alternative [29]. Tensoring nonlinear decoders spread errors across logical coordinates, defeating amplification.

Comparisons with Ellis–Keller–Lifshitz edge-isoperimetric stability [27] and Gowers–Hatami stability of approximate representations [30] did not yield the desired decoder. Graphical encodings offered local consistency but required exponentially many Fourier queries. Håstad’s parity gap [32] and Rao’s repetition [51] supplied a proposed hardness route. The assistant questioned expander links against Bafna and collaborators’ sum-of-squares algorithms [5]. It also noted that one label determines a connected perfectly satisfied unique-game component.

To coordinate scalar errors in nearly satisfiable affine games, it considered combining Goemans–Williamson rounding with Blum–Luby–Rubinfeld linearity testing [29, 13]. Random hashing invoked Charikar–Makarychev–Makarychev’s smaller-alphabet guarantee [17], but incompatible projected solutions prevented a geometric-progress argument. It contrasted polynomial time with Arora–Barak–Steurer’s subexponential algorithm and questioned the hierarchy strength of Khot–Saket gaps [1, 43]. Its proposed use of Karger’s cut-counting bound involved a potentially unbounded approximation factor [35]; Friedgut’s cube junta theorem lacked the needed general-graph analogue [28].

Independently controlled transposition words failed to generate the required transitive period action. Barrington’s theorem suggested both an expressivity obstruction and an escape through repeated control bits [10].

The assistant considered Croot–Lev–Pach polynomial methods and Lovett’s analytic-rank work, but found its contemplated inverse-rank estimate too weak [20, 45]. It then set $X = K = \mathbb{F}_q^3$, $q = 2^d$, and

$$C(x, y) = y + (x_2x_3, x_1x_3, x_1x_2).$$

It compared nonlinear error p with compatible linear-decoder error h , arguing that cubic monomials and finite-field specialization bound exceptional cancellations by $3r$ for rank- r observable families. Concatenating through all binary-linear isomorphisms randomized observable subspaces while enforcing a common parent lift. The assistant argued that its harmonic-potential estimate kept $h \geq 1/4$ while p vanished; the assistant checked specialization and conditioning.

To convert this separation into polynomial-size hardness, it returned to Khot–Minzer–Safra soundness through Barak–Kothari–Steurer [39, 8], the Dinur–Khot–Kindler–Minzer–Safra agreement framework and its smoothness/query-size obstacle [21], and Minzer–Zheng’s *Near Optimal Alphabet-Soundness Tradeoff PCPs* [46]. Outer-PCP composition remained its next task.

6 Shared affine tables and frequent decoding advice

The original BasicSDP threshold problem now motivates a proposed route through Unique Games hardness. The assistant combines its nonlinear encoding with shared affine tables, then develops a transfer to finite constraint languages.

The encoding seeks an equivariant decoder on a binary space containing $K = \mathbb{F}_2^\ell$: carefully chosen noise should rarely change the decoded value, while every sufficiently high-rank family of linear characters detects it with constant probability. Concatenated quadratic constructions and a harmonic rank potential are offered to establish this separation.

The assistant studies subspace identifications in Minzer–Zheng’s PCP construction [46] and clique folding in the Dinur–Khot–Kindler–Minzer–Safra reduction [21]. The proposed reduction replaces approximate consistency checks by exact identification of tables with the same reduced support. This matters because consistency errors could otherwise concentrate on the small slices used by the published

shortcode soundness results of Khot–Minzer–Safra and Barak–Kothari–Steurer [39, 8]. Conditional agreement suggests decoding an affine class rather than enumerating individual candidates. Fourier sampling is proposed to recover a compatible projected label.

A further obstacle is that existence of one favorable slice does not guarantee useful random advice:

VERBATIM EXCERPT

Existence tiny row fiber via theorem not enough because advice U gets random $A, S=AM$. Need positive probability independent k of hitting a whole fiber.

The proposed repair randomizes the union of favorable row fibers and argues that too little total measure would contradict shortcode soundness. Starting from Håstad’s parity-equation hardness theorem [32], sparse projections and coordinates with zero advice slopes are intended to expose repeated equation-versus-variable games.

For BasicSDP, the assistant uses Raghavendra’s dictatorship-test framework [50]: smoothing and Gaussian comparison aim to bound low-influence test performance by the integral optimum; influential coordinates instead decode the outer game. It develops a direct Lindeberg comparison instead of invoking the Mossel–O’Donnell–Oleszkiewicz invariance theorem [47]. The assistant identifies two remaining concerns: repeated variables may obstruct the intended projection-game formulation, and ordinary Raz repetition [53] introduces an alphabet dependence that risks circular parameter choices. Alphabet-independent projection-game repetition from Rao and Dinur–Steurer [51, 22] is checked as a remedy.

The assistant also compares the proposed parameters with the alphabet-dependent guarantees of Charikar–Makarychev–Makarychev [17], finding no contradiction in that comparison. It revisits a fall-back converse based on the Khot–Vishnoi integrality gap [38], extended with dummy labels to force a baseline objective value.

7 Uniformity in rectangular shortcode parameters

Combining the nonlinear gadget with Unique Games decoding requires bounds that remain uniform as the outer construction grows.

The assistant asks whether translation constraints over $\mathbb{F}_2^\ell$ admit an algorithmic obstruction. It revisits Khot’s Unique Games formulation [44] and recalls the Max-2-Lin hardness results of Khot–Kindler–Mossel–O’Donnell [40], while questioning their relevance to this particular group. Coordinatewise Goemans–Williamson rounding [29] incurs a union-bound loss in ℓ ; the alphabet dependence in the Charikar–Makarychev–Makarychev algorithms [17] also enters the comparison. The entangled-prover result *Unique Games with Entangled Provers are Easy* [37] does not directly give an algorithm for the classical game.

The gadget has an equivariant decoder $C : \mathcal{U} \rightarrow K$: shifting its input by $k \in K$ shifts its output by k . Proposed noise almost preserves this nonlinear decoder while remaining detectable by sufficiently large families of linear characters. A finite-field quadratic map, a bound of $3r$ simultaneous alignments for generic rank- r character spaces, and concatenation tracked by a harmonic potential are used by the assistant to argue for the separation. The assistant checks why specialization cannot increase matrix rank:

VERBATIM EXCERPT

Matrix rank cannot rise under specialization because all larger minors zero *as polynomials*, even with rational combinations; rank over function field r implies determinants identically zero in $\mathbb{F}_2[x]$. yes. Good.

The assistant uses the Khot–Vishnoi integrality gap [38] to check simple SDP rounding objections. It invokes the Khot–Minzer–Safra inverse result [39], consulted through Barak–Kothari–Steurer’s short-code formulation [8]. It scrutinizes whether the constants remain uniform for the construction’s highly rectangular matrices and provisionally accepts that they do.

For soundness, the assistant uses Håstad’s parity-equation hardness theorem [32] and checks a cloning modification to make the three variables within each equation distinct. Ordered local copies preserve compatibility without merging repeated variable positions across blocks. It credits Minzer–Zheng [46] for the smooth variable-versus-equation game with advice that informs its sparse-projection construction. Sparse singleton projections, with probability $\beta = k^{-2/3}$, are intended both to make distributional error vanish and to leave increasingly many coordinates available for decoding. Fourier agreement, bounded row and column advice, and projection-game repetition must then rule out cheating. The assistant notes that conditioning on successful witnesses would compromise the independence needed for decoding. To avoid circular parameter choices, the assistant compares the alphabet-dependent repetition bounds of Raz and Holenstein [53, 33] with the alphabet-independent projection-game bounds of Rao and Dinur–Steurer [51, 22].

The BasicSDP transfer follows Raghavendra’s framework [50], using shared answers for repeated variables, smoothing, low-influence Gaussian comparison based on pair moments, and influential-coordinate decoding. Rational approximations require reserved slack.

8 Row-fiber abundance and threshold transfer

The next issue is whether useful decoding advice occurs with constant probability. The proposed argument invokes Håstad’s parity-equation hardness result [32], the Khot–Minzer–Safra shortcode theorem in the Barak–Kothari–Steurer formulation [39, 8], and projection-game parallel repetition [51, 22]; their applicability depends on the uniformity and conditioning requirements below.

The proposed gadget separates a shift-equivariant nonlinear map’s small noise sensitivity from constant detection by sufficiently high-rank linear observables. Its quadratic seed, $Q(x) = (x_2x_3, x_1x_3, x_1x_2)$, operates over a characteristic-two field. A polynomial anti-alignment argument and harmonic rank-loss potential were intended to amplify that separation. The argument must account for random orientations, quotient shifts, and possible representations of nonlinear derivatives by mixtures of linear maps before controlling the frequency of useful decoding advice.

Soundness exposed a sharper issue: the shortcode theorem’s existence of a correlated slice did not immediately make useful row advice sufficiently frequent.

VERBATIM EXCERPT

Row fiber abundance lemma is important to avoid conditioned advice rare due row values S_0 exponential m . KMS just existence of one fiber, probability $2^{-m/r}$. Need union randomization argument ensure many.

The proposed repair randomized the union of good row fibers to argue abundance by contradiction. Another correction made the goodness probability explicitly averaged over questions, rather than valid for each question. The sparse-projection and matrix-advice construction drew on the smooth variable-versus-equation game with advice of Minzer and Zheng [46]. The assistant argues that sparse projection at rate $\beta = k^{-2/3}$ gives vanishing distributional error while retaining many coordinates with zero advice slopes. Applying parallel repetition then depended on those hidden coordinates remaining independent under public conditioning. For this fixed-alphabet outer game, the assistant also considered the general repetition theorems of Raz and Holenstein [53, 33]; alphabet-independent amplification remained the role of Rao or Dinur–Steurer [51, 22].

The BasicSDP transfer followed Raghavendra’s dictatorship-test framework [50] and used a finite gap instance with integral optimum A_* , feasible SDP value B_* , and $A_*/B_* < r$. Smoothing, Fourier truncation, and Gaussian replacement were intended to bound low-influence acceptance by A_* ; influential coordinates supplied candidate Unique Games labels. Parameter ordering and rational approximation also received scrutiny. For comparison, the assistant considered applying Goemans–Williamson rounding separately to binary coordinates [29]; its union-bound estimate accumulated a factor equal to the number of coordinates, so this did not supply the desired alphabet-independent obstruction.

9 Scalar detection and conditional independence

The relation between scalar noise cost and simultaneous linear decoding becomes the next obstacle. The proposed reduction continues to use Håstad’s parity-equation gap [32], the Khot–Minzer–Safra inverse shortcode result [39] through Barak–Kothari–Steurer’s *Small-Set Expansion in Shortcode Graph and the 2-to-2 Conjecture* [8], projection-game parallel repetition [22], and Raghavendra’s BasicSDP framework [50]. In checking that the starting gap reduction remains deterministic and polynomial for fixed parameters, it invokes the PCP constructions of Dinur [23] and Arora–Safra [3].

The gadget seeks a perturbation that rarely changes an equivariant nonlinear encoding but remains detectable by high-rank linear observations. The algebraic argument identifies which monomial evaluations generate a function space; the recursive argument seeks to control exceptional rank losses through a harmonic estimate. Searching for an obstruction, the assistant observes:

VERBATIM EXCERPT

Aha! We don't actually need simultaneous union per z ; detection prob for subspace G equals $\frac{1}{2}$ times average over its characters of scalar nonzero probability (because for each a with some $g \neq 0$, exactly half of linear subspace maps a to 1).

The assistant thus reformulates the possible contradiction through average scalar cost, but still needs compatible linear extensions to complete this obstruction to the nonlinear separation. It also tests the gadget against the Kahn–Kalai–Linial influence theorem [34]: small average influence across an enormous number of leaves does not give the sought contradiction. Comparing binary translation games with the Goemans–Williamson near-satisfiable Max-Cut guarantee [29] likewise leaves the difficulty of controlling simultaneous errors across all label coordinates.

The smooth variable-versus-equation game with advice in Minzer–Zheng [46] supplies the inspiration for sparse projections and zero-advice coordinates. Decoding imposes two consistency requirements. Shared affine tables must choose translation representatives from their reduced keys alone. Also, bounded row information must live in the fixed output dimension, while column witnesses may be selected using the equation prover’s information; blindly guessing directions in a growing domain would destroy a constant decoding probability. The assistant argues that sparse projection with probability $\beta = k^{-2/3}$ makes advice distributions close while leaving increasingly many coordinates with zero advice slopes. Conditional independence on those coordinates remains essential to the repetition argument. The assistant considers Raz’s general repetition theorem [53] for the fixed-alphabet clean game, while using the alphabet-independent projection-game bounds of Rao and Dinur–Steurer [51, 22] for the final amplification.

An alternative amplification using several perturbations is explored and left unused. The retained BasicSDP transfer smooths and truncates functions, matches singleton and pair moments with Gaussian variables, and requires one assignment across the gap instance and alphabet-independent influence bounds.

10 Compatibility of labels and pair moments

Local decoding must yield compatible labels before the Unique Games gap can be transferred to a fixed constraint language.

The central proposal separates nonlinear stability from linear detectability. A map $C : \mathcal{U} \rightarrow K$ obeys $C(x + k) = C(x) + k$; carefully chosen noise should rarely change C while detecting every sufficiently high-rank family of characters restricted to K . The recursive quadratic gadget uses

$$Q(x) = (x_2x_3, x_1x_3, x_1x_2), \quad x \in \mathbb{F}_{2^d}^3.$$

A generic-subspace argument is intended to limit aligned characters, and a harmonic rank potential should control extinction while nonlinear sensitivity decreases. Attempts to contradict this separation through coordinatewise linear rounding encounter a difficulty: separately corrected coordinates need not remain in the required linear code.

The outer reduction starts with Håstad’s parity-triple hardness [32]. Sparse projections and matrix advice follow the smooth-game idea discussed by Minzer and Zheng [46]. The assistant checks repetition on the clean coordinates using Raz’s theorem [53], and alphabet-independent amplification using Rao or Dinur–Steurer [51, 22].

The proposed soundness bridge invokes the shortcode results of Khot–Minzer–Safra and Barak–Kothari–Steurer [39, 8]. The proposed application decodes on bounded row and column slices and transfers useful events by unconditional total-variation comparison. Folding should select valid affine answers; coordinates with zero advice should recover an ordinary repeated projection game. Repeated variables retain separate local coordinates, a convention essential to the claimed independence.

The assistant also considers algorithms for Unique Games on expanding constraint graphs [2], but argues that its graph need not have the required expansion. It considers coordinatewise Boolean rounding associated with Goemans–Williamson and Charikar–Wirth [29, 18]; a union bound over many coordinates does not give the desired alphabet-independent guarantee. It compares translation-game examples from Khot–Vishnoi [38] with the proposed construction, distinguishing SDP stability from stability of an actual labeling.

Finally, the transfer to the BasicSDP threshold in Raghavendra’s framework [50] is checked for matched pair moments, Gaussian replacement after smoothing and truncation, and simplex projection producing one globally consistent assignment across constraints. Rational weights and fixed-parameter ordering are also revisited.

11 Parameter order and uniform slice bounds

The order of parameter choices is essential to the proposed route from Håstad’s parity-equation gap [32] through a nonlinear noise gadget and Unique Games hardness to each fixed finite constraint language. The gadget aims to make an equivariant decoder nearly invariant under noise while high-rank linear observables detect that noise with constant probability. Rechecking the finite-field construction leads through independence of square-root monomials, polynomial specialization, and a harmonic potential controlling rank loss under recursive restrictions.

A concrete counterstrategy assigns fixed bits on small table supports and chooses satisfying local answers elsewhere. The assistant argues that simultaneous perturbation directions impose incompatible demands on unsatisfied equations, while uniform subspace noise obstructs concentrating on favorable directions.

The assistant next checks a literature dependency: the shortcode result attributed to Khot–Minzer–Safra through the Barak–Kothari–Steurer formulation [39, 8] must provide slice bounds independent of

a subsequently enlarged output dimension. Otherwise the row constraints might consume the kernel needed for decoding. The assistant considers rectangular padding and projection as possible ways to obtain the required uniform bounds.

The soundness audit draws on the smooth variable-versus-equation game with advice in Minzer and Zheng’s *Near Optimal Alphabet-Soundness Tradeoff PCPs* [46], and examines whether zero advice slopes conceal singleton projections. The assistant argues that with projection probability $\beta = k^{-2/3}$, distributional disturbance vanishes while the number of usable repeated-game coordinates grows. Decoder dependencies and parameter order are clarified. The assistant checks the alphabet-independent repetition bounds of Rao and Dinur–Steurer [51, 22] for amplifying the resulting Unique Games gap; it also notes that Raz’s theorem [53] suffices for the fixed-alphabet game on the clean coordinates. An alternative amplification by summing perturbations remains unused because inconsistent loops require additional control.

For comparison with known algorithms, it revisits the Khot–Vishnoi integrality gap [38]: the binary translation structure of the constraints does not by itself supply a dimension-independent SDP rounding argument.

Finally, the assistant checks the BasicSDP transfer in Raghavendra’s framework [50] through full-support marginals, smoothing, Gaussian moment matching, and influence-list decoding. The dimension dependence of the imported shortcode bounds remains under examination.

12 Assembling the proposed reduction

The proposed reduction has two stages: first obtain Unique Games with nearly perfect completeness and arbitrarily small soundness, then transfer that gap to the BasicSDP threshold for each fixed finite constraint language.

The quadratic gadget is checked through a harmonic potential measuring surviving character rank. A key local observation is that nongeneric character spaces need only have small constant probability: fresh outgoing randomness supplies a further factor proportional to the loss parameter. This step also requires trace duality, character alignment, and conditional uniformity of gadget outputs.

The assistant uses the inverse-shortcode results of Khot–Minzer–Safra and the Barak–Kothari–Steurer formulation [39, 8]. It explains an equality-test consequence through value fibers and justifies dimension padding by projecting affine tensor-product slices.

The parity-triple reduction starts from Håstad’s nearly satisfiable parity-equation hardness theorem [32]. The assistant credits Minzer and Zheng’s PCP construction as inspiration for the variable-versus-equation game with advice [46]. Its proposed soundness analysis compares gadget noise with uniform rank-one noise, then seeks affine structure on bounded row-and-column slices. Its parameter order fixes gadget and advice dimensions before increasing the number k of parity blocks. Sparse projection with $\beta = k^{-2/3}$ is intended to make $k\beta^2 \rightarrow 0$ while $k\beta \rightarrow \infty$, simultaneously suppressing distributional error and producing many clean singleton blocks. Uniform bounds and alphabet-independent projection-game repetition, supplied by Rao or Dinur–Steurer, are essential dependencies [51, 22]. The assistant also considers an alternative amplification using Raz’s theorem on a small base game, but retains the projection-game route [53].

Finally, using Raghavendra’s SDP and dictatorship-test framework, the assistant transfers a fixed-language SDP gap witness through a smoothed dictatorship test, low-influence replacement, list decoding, and rational sampling weights [50]. It also compares the claimed threshold with Goemans–Williamson approximation for Max-Cut and finds no contradiction [29].

The assistant concludes with its proposed affirmative argument for the Unique Games consequence and the BasicSDP threshold claim.

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